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3.2 Particles and Radiation

1. Atomic Structure

Atom: A nucleus (protons + neutrons) surrounded by orbiting electrons. The nucleus contains almost all the mass but occupies a tiny fraction of the volume (~10^{-15} m diameter vs ~10^{-10} m for the atom). The nucleus is therefore approximately 10^5 times smaller than the atom.

Particle	Relative Charge	Relative Mass	Absolute Charge / C	Rest Mass / kg
Proton	+1	1	$+1.60 \times 10^{-19}$	1.673×10^{-27}
Neutron	0	1	0	1.675×10^{-27}
Electron	-1	1/1836	-1.60×10^{-19}	9.109×10^{-31}

Notation

A nuclide is written as $^{A}_{Z}X$, where A = mass number (protons + neutrons), Z = atomic number (protons). The number of neutrons N = A - Z.

Isotopes: Atoms of the same element (same Z) with different numbers of neutrons (different A). They have identical chemical properties (same electron configuration) but different physical properties (e.g. density, stability, nuclear behaviour).

Real-world application: Isotopes have widespread medical and industrial uses. Carbon-14 is used in radiocarbon dating of archaeological artefacts. Iodine-131 is used to treat thyroid cancer. Uranium-235 and Uranium-238 are isotopes with very different nuclear properties — only U-235 undergoes thermal neutron fission.

Specific Charge

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$$\text{Specific charge} = \text{charge / mass (units: C kg}^{-1}\text{)}$$

Specific charge is useful for identifying particles in mass spectrometers and particle detectors, because particles with different specific charges follow different curved paths in electric and magnetic fields.

Worked Example: Calculate the specific charge of (a) a proton, (b) an electron, (c) a He-4 nucleus.

(a) Proton: specific charge = $1.60 \times 10^{-19} / 1.673 \times 10^{-27}$ = **9.56×10^{7} C kg^{-1}**

(b) Electron: specific charge = $1.60 \times 10^{-19} / 9.109 \times 10^{-31}$ = **1.76×10^{11} C kg^{-1}**
(The electron has a much higher specific charge due to its tiny mass)

(c) He-4 nucleus (alpha particle): charge = $2 \times 1.60 \times 10^{-19} = 3.20 \times 10^{-19}$ C
mass = $4 \times 1.673 \times 10^{-27} = 6.692 \times 10^{-27}$ kg
specific charge = $3.20 \times 10^{-19} / 6.692 \times 10^{-27}$ = **4.78×10^{7} C kg^{-1}**

Exam Tip: When calculating specific charge, be careful to use the total charge and total mass of the particle or ion. For ions, include the charge from missing/extra electrons. For example, a Na^{+} ion has charge +e but mass approximately 23u.

2. Stable and Unstable Nuclei

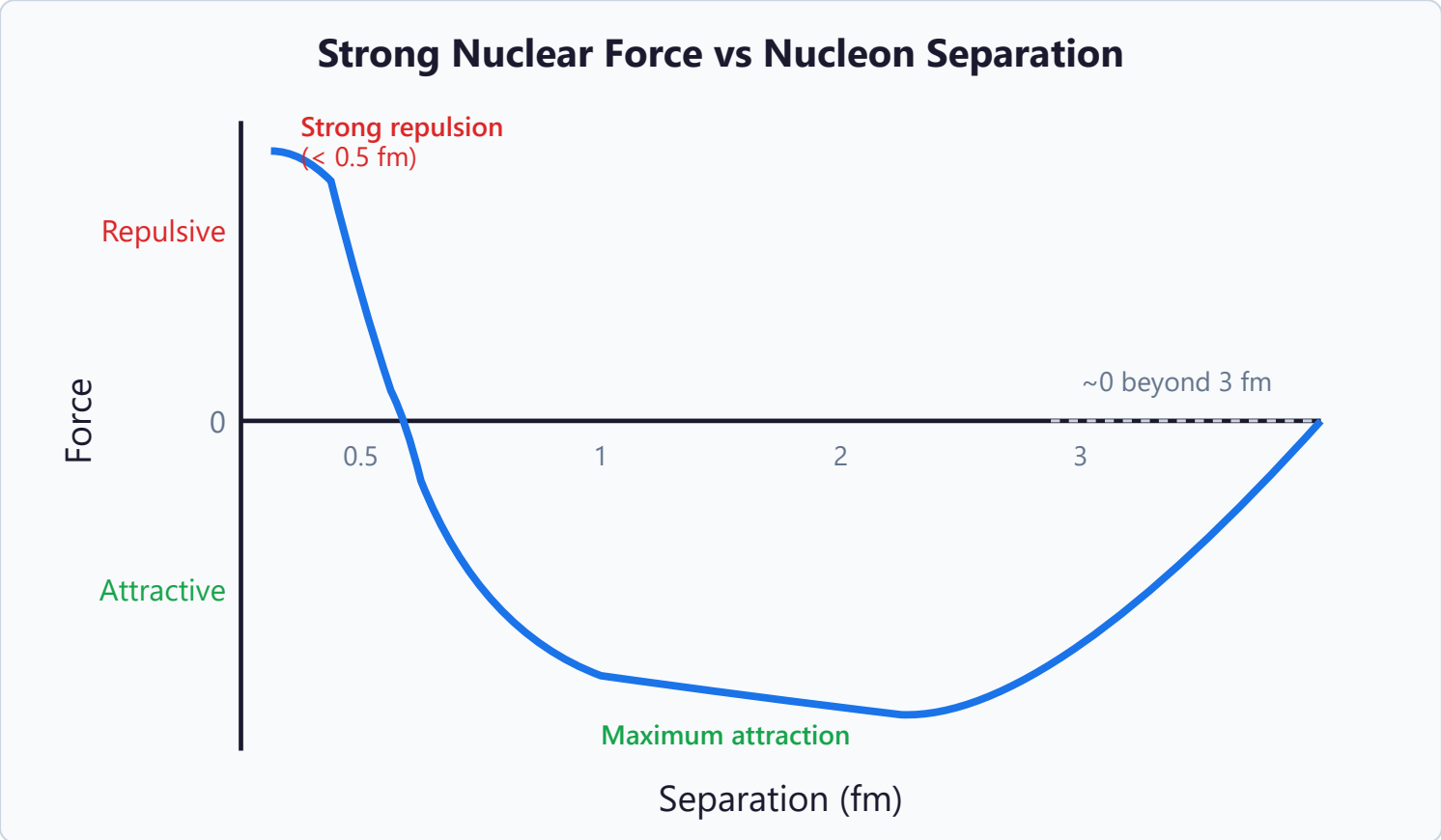
The strong nuclear force holds the nucleus together, overcoming electrostatic repulsion between protons. Nuclei are unstable if they have:

- Too many neutrons (neutron-rich) → beta-minus decay
- Too many protons (proton-rich) → beta-plus decay or electron capture
- Too many nucleons overall (too heavy, A > ~82) → alpha decay

The Strong Nuclear Force

- **Range:** Very short — effective only up to about 3-4 fm (femtometres).
- **Attractive** from about 3 fm down to about 0.5 fm — this holds the nucleus together.
- Becomes **repulsive** below about 0.5 fm (prevents nucleons collapsing into each other).

- Has zero effect beyond about 3-4 fm — this is why very large nuclei are less stable.
- Acts between **all nucleons** (p-p, p-n, n-n) equally — it is charge-independent.
- At its effective range, it is much stronger than the electromagnetic force.



Types of Radioactive Decay

Decay	Particle Emitted	Change in A	Change in Z	Quark Change
Alpha (α)	$^{4}_{2}He$ nucleus	-4	-2	N/A
Beta-minus (β^-)	Electron + antineutrino	0	+1	d → u
Beta-plus (β^+)	Positron + neutrino	0	-1	u → d

In β^- decay: $n \rightarrow p + e^- + \bar{\nu}_e$

In β^+ decay: $p \rightarrow n + e^+ + \nu_e$

In electron capture: $p + e^- \rightarrow n + \nu_e$

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Exam Tip: Beta-plus decay and electron capture both reduce the proton number by 1 and increase the neutron number by 1. They are alternative processes for proton-rich nuclei. Electron capture is more common in heavier nuclei where inner orbital electrons are closer to the nucleus.

3. The Four Fundamental Forces

Force	Range	Acts On	Exchange Particle	Relative Strength
Strong nuclear	~3-4 fm	Hadrons (quarks)	Gluon (g)	1
Electromagnetic	Infinite ($1/r^2$)	Charged particles	Virtual photon (γ)	$\sim 10^{-2}$
Weak nuclear	$\sim 10^{-18}$ m	All fermions	W^+, W^-, Z^0 bosons	$\sim 10^{-6}$
Gravitational	Infinite ($1/r^2$)	All particles with mass	Graviton (theoretical)	$\sim 10^{-39}$

Exchange particles (gauge bosons): Forces between particles are mediated by the exchange of virtual particles. When two particles interact, they exchange a gauge boson, which transfers momentum and energy between them. This is the quantum mechanical model of forces.

Why the weak force has short range: The W and Z bosons are very massive (W: ~80 GeV/c^2, Z: ~91 GeV/c^2). By the Heisenberg uncertainty principle, such massive virtual particles can only exist for an extremely brief time, limiting the range to $\sim 10^{-18}$ m.

Real-world application: The weak force is responsible for beta decay, which powers the proton-proton chain in the Sun. Without the weak force, hydrogen could not fuse into helium, and stars would not shine. The weak force also causes neutrino interactions used in neutrino detectors deep underground.

4. Classification of Particles

4.1 Hadrons

Hadrons: Particles that experience the strong nuclear force. They are made of quarks. Hadrons are divided into baryons (3 quarks) and mesons (quark-antiquark pair).

Type	Quark Composition	Baryon Number	Examples
Baryons	3 quarks (qqq)	+1	Proton (uud), Neutron (udd), Sigma, Lambda, Xi
Antibaryons	3 antiquarks ($\overline{qqq}$)	-1	Antiproton ($\overline{uud}$)
Mesons	Quark-antiquark ($q\bar{q}$)	0	Pion (π^+, π^0, π^-), Kaon (K^+, K^0, K^-)

The proton is the only stable baryon. All other baryons eventually decay into protons. The neutron is unstable when free (half-life ~10 minutes) but stable inside most nuclei.

4.2 Leptons

Leptons: Fundamental particles that do NOT experience the strong force. They have a lepton number of +1 (antileptons have -1). Leptons are not made of quarks — they are fundamental.

Generation	Charged Lepton	Neutrino
1st	Electron (e^-)	Electron neutrino (ν_e)
2nd	Muon (μ^-)	Muon neutrino (ν_μ)
3rd	Tau (τ^-)	Tau neutrino (ν_τ)

Each generation has its own lepton number (electron lepton number L_e, muon lepton number L_μ) which is separately conserved in all interactions.

Real-world application: Muons are produced when cosmic rays hit the upper atmosphere. Despite their short lifetime (~2.2 μs), they reach the Earth's surface due to time dilation predicted by special relativity. This was one of the first experimental confirmations of Einstein's theory.

Standard Model — Particle Classification

QUARKS
Feel strong force

Gen 1	Gen 2	Gen 3
u (+2/3)	c (+2/3)	t (+2/3)
d (-1/3)	s (-1/3)	b (-1/3)

Baryons = qqq
Mesons = qq
(together = Hadrons)
HADRONS

LEPTONS
Do NOT feel strong force

Gen 1	Gen 2	Gen 3
e-	mu-	tau-
ve	v mu	v tau

Lepton number = +1
Antileptons: L = -1
Fundamental particles

EXCHANGE BOSONS
Force carriers

Gluon	Strong
Photon (y)	EM
W+, W-, Z0	Weak
Graviton*	Gravity

*theoretical, not yet observed

Key AQA Quarks: Up (u) +2/3, Down (d) -1/3, Strange (s) -1/3
Antiquarks have opposite charge, baryon number, and strangeness
Baryon number: quarks = +1/3, antiquarks = -1/3

5. Quarks

AQA requires knowledge of three quark flavours: up, down, and strange. Charm, top, and bottom exist but are not required.

Quark	Symbol	Charge (e)	Baryon Number	Strangeness
Up	u	+2/3	+1/3	0
Down	d	-1/3	+1/3	0
Strange	s	-1/3	+1/3	-1

Antiquarks ($\bar{u}$, $\bar{d}$, $\bar{s}$) have opposite charge, baryon number, and strangeness to their quark counterparts.

Checking Quark Compositions

Worked Example: Verify the quark compositions and properties of key particles.

Proton (uud):

Charge = +2/3 + 2/3 - 1/3 = +1 ✓
Baryon number = +1/3 + 1/3 + 1/3 = +1 ✓
Strangeness = 0 + 0 + 0 = 0 ✓

Neutron (udd):
Charge = +2/3 - 1/3 - 1/3 = 0 ✓
Baryon number = +1/3 + 1/3 + 1/3 = +1 ✓

π^+ pion ($u\bar{d}$):
Charge = +2/3 + 1/3 = +1 ✓ (the antiquark $\bar{d}$ has charge +1/3)
Baryon number = +1/3 - 1/3 = 0 ✓ (mesons have B = 0)

K^+ kaon ($u\bar{s}$):
Charge = +2/3 + 1/3 = +1 ✓
Strangeness = 0 + (+1) = +1 ✓ (anti-strange has strangeness +1)

Key Meson Quark Compositions

Meson	Quark Content	Charge	Strangeness
π^+	$u\bar{d}$	+1	0
π^-	$d\bar{u}$	-1	0
π^0	$u\bar{u}$ or $d\bar{d}$	0	0
K^+	$u\bar{s}$	+1	+1
K^-	$s\bar{u}$	-1	-1
K^0	$d\bar{s}$	0	+1

Exam Tip: Strangeness is conserved in strong interactions but can change by 0 or ±1 in weak interactions. If a kaon (strangeness = ±1) is produced in a strong interaction, another strange particle must be produced alongside it (associated production) to conserve strangeness. Kaons decay via the weak interaction because strangeness changes.

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6. Antiparticles

Antiparticle: Every particle has a corresponding antiparticle with the same mass and rest energy but opposite charge, baryon number, lepton number, and strangeness. Predicted by Dirac (1928) and confirmed by Anderson's discovery of the positron (1932).

Particle	Antiparticle	Notes
Electron (e^-)	Positron (e^+)	First antiparticle discovered (Anderson, 1932)
Proton (p)	Antiproton ($\bar{p}$)	Charge = -1, baryon number = -1
Neutron (n)	Antineutron ($\bar{n}$)	Same charge (0) but opposite baryon number
Neutrino (ν_e)	Antineutrino ($\bar{\nu}_e$)	Opposite lepton number

Pair Production and Annihilation

Annihilation: A particle meets its antiparticle, both are destroyed, and their mass-energy is converted into two gamma photons (two photons are needed to conserve momentum).

$$\text{Minimum energy of each photon: } E_{photon} = m_0c^2$$

Worked Example: Calculate the minimum energy and wavelength of photons produced when an electron and positron annihilate at rest.

Rest energy of electron = $m_ec^2 = 9.109 \times 10^{-31} \times (3.0 \times 10^8)^2 = 8.20 \times 10^{-14}$ J = 0.511 MeV

Each photon carries at least this energy: E = 0.511 MeV = 8.20×10^{-14} J

Wavelength: $\lambda = hc/E = (6.63 \times 10^{-34} \times 3.0 \times 10^8) / 8.20 \times 10^{-14}$ = **2.43×10^{-12} m**
This is in the gamma ray region of the EM spectrum.

Pair production: A photon with sufficient energy creates a particle-antiparticle pair. The photon must have energy of at least $2m_0c^2$. This usually occurs near a nucleus (to conserve momentum — the nucleus recoils to absorb the excess momentum).

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Worked Example: Calculate the minimum photon energy for electron-positron pair production.

$E_{min} = 2m_ec^2 = 2 \times 0.511$ MeV = **1.022 MeV**

The corresponding minimum frequency: $f = E/h = 1.022 \times 10^6 \times 1.6 \times 10^{-19} / 6.63 \times 10^{-34} = 2.47 \times 10^{20}$ Hz (gamma ray)

Real-world application: PET (Positron Emission Tomography) scanners exploit electron-positron annihilation. A radioactive tracer emits positrons which annihilate with electrons in the body, producing pairs of gamma photons detected by the scanner. This is used to image brain activity, detect cancers, and monitor heart function.

7. Particle Interactions and Conservation Laws

Conservation Laws

In all interactions, the following quantities are conserved:

Quantity	Conserved in Strong?	Conserved in Weak?	Conserved in EM?
Energy (including mass-energy)	Yes	Yes	Yes
Momentum	Yes	Yes	Yes
Charge	Yes	Yes	Yes
Baryon number	Yes	Yes	Yes
Lepton number (each flavour)	Yes	Yes	Yes
Strangeness	Yes	Can change by ±1	Yes

Worked Example: Verify that beta-minus decay conserves all quantum numbers.

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$n \rightarrow p + e^- + \bar{\nu}_e$

Charge: $0 \rightarrow +1 + (-1) + 0 = 0$ ✓
Baryon number: $+1 \rightarrow +1 + 0 + 0 = +1$ ✓
Electron lepton number: $0 \rightarrow 0 + 1 + (-1) = 0$ ✓
(The antineutrino has $L_e = -1$, the electron has $L_e = +1$)

All conserved. This is a weak interaction (W^- boson mediates it).

Key Interactions at A-Level

Beta-minus decay (weak interaction):

$n \rightarrow p + e^- + \bar{\nu}_e$

At the quark level: $d \rightarrow u + e^- + \bar{\nu}_e$ (mediated by W^- boson)

Beta-plus decay (weak interaction):

$p \rightarrow n + e^+ + \nu_e$

At the quark level: $u \rightarrow d + e^+ + \nu_e$ (mediated by W^+ boson)

Electron capture (weak interaction):

$p + e^- \rightarrow n + \nu_e$ (mediated by W^+ boson)

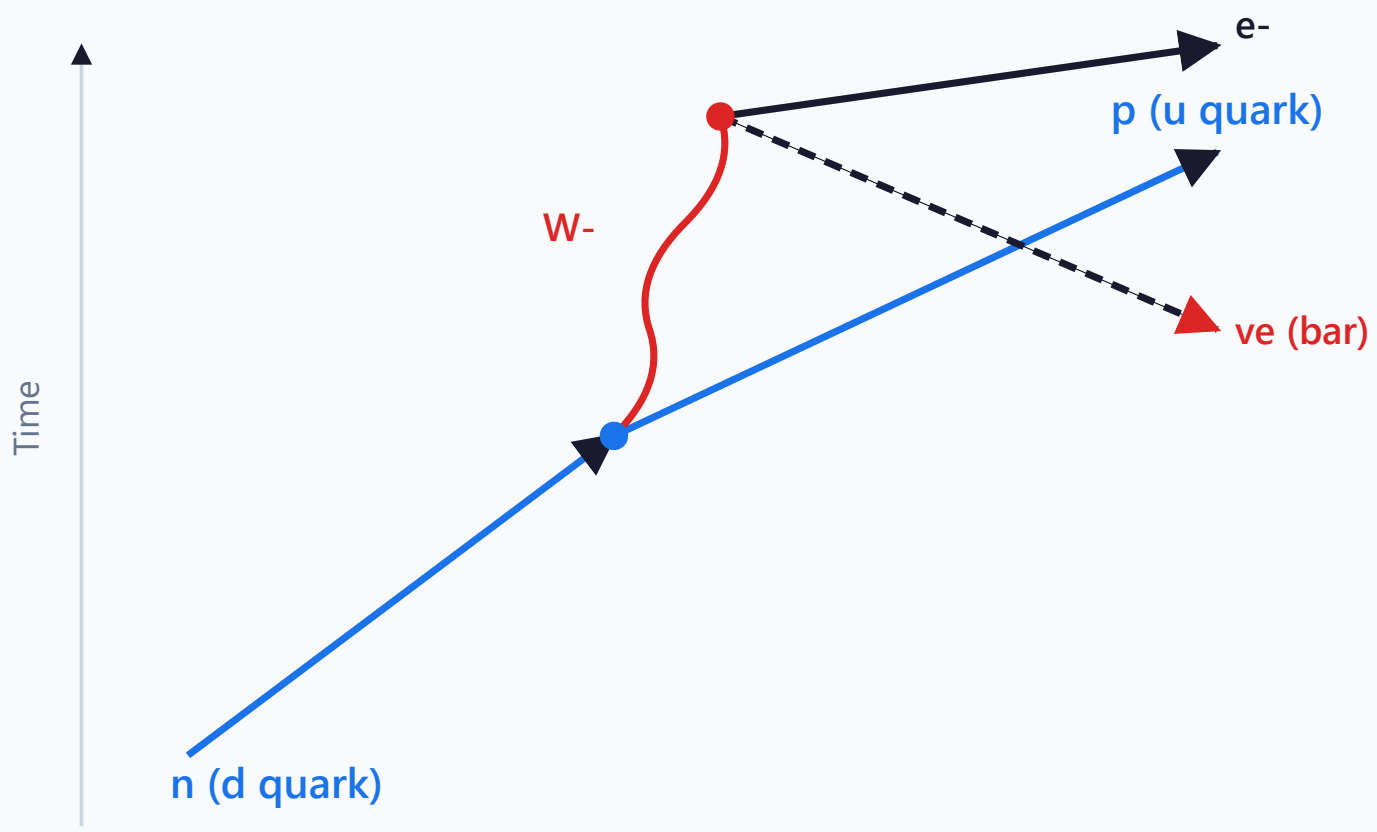
Electron-proton collision (weak interaction, if energy is sufficient):

$p + e^- \rightarrow n + \nu_e$

8. Feynman Diagrams

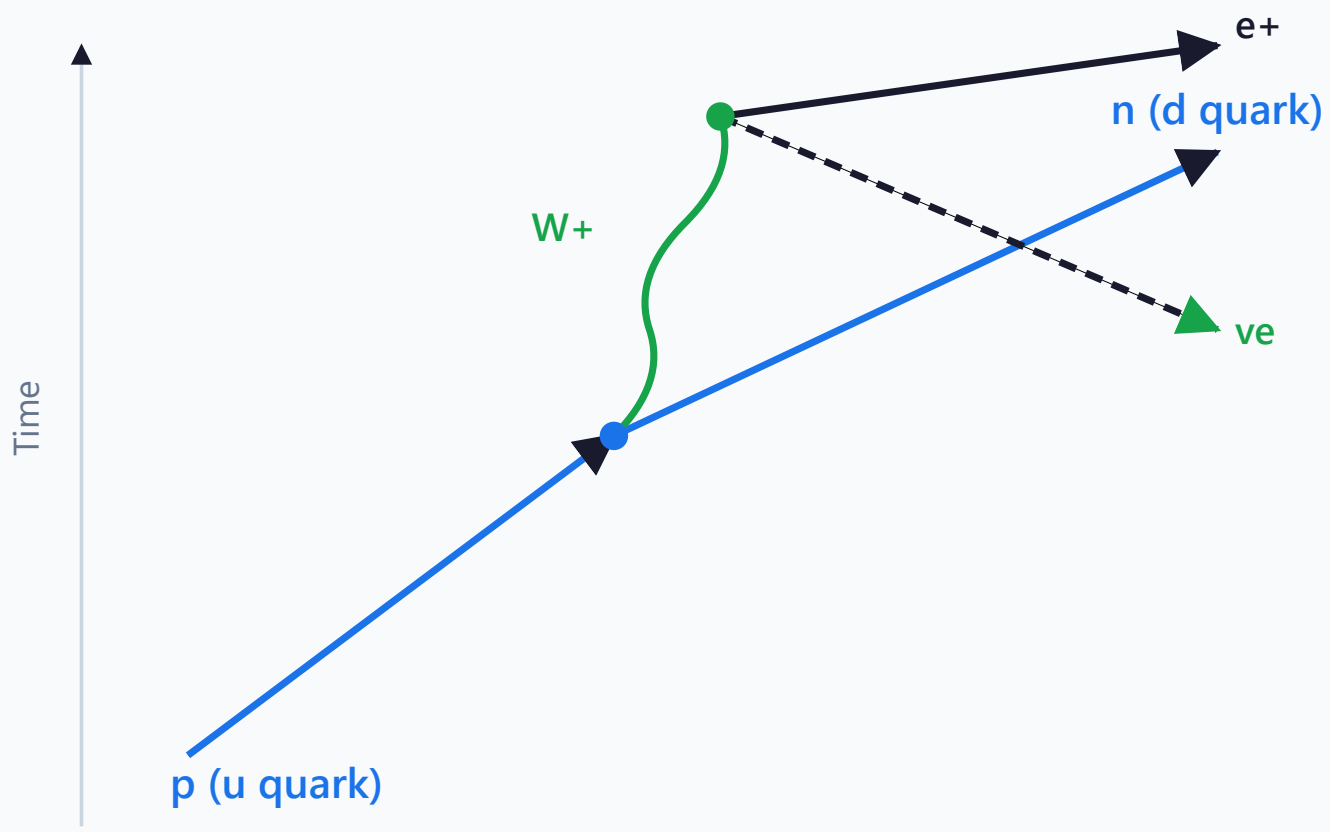
Feynman diagrams are pictorial representations of particle interactions. Time runs upward (or left-to-right). Straight lines represent fermions (quarks, leptons). Wavy or curly lines represent exchange bosons.

Feynman Diagram: Beta-minus Decay

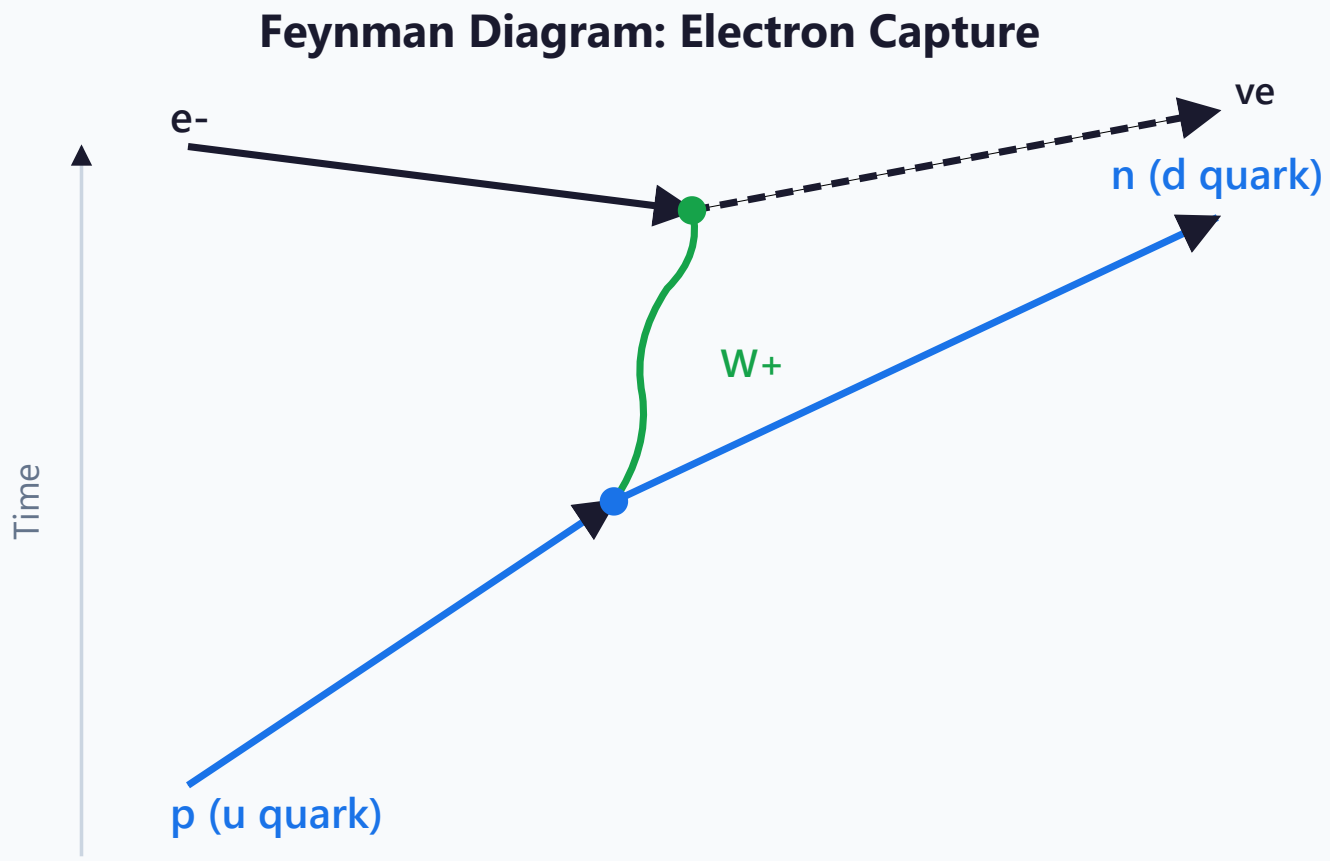


d quark emits W- boson, becomes u quark. W- decays into e- and antineutrino.

Feynman Diagram: Beta-plus Decay



u quark emits W+ boson, becomes d quark. W+ decays into e+ and neutrino.



Proton absorbs electron via W+ exchange, producing neutron + neutrino.

Exam Tip: In Feynman diagrams, W bosons change quark flavour. W^- emission turns d → u. W^+ emission turns u → d. The W then decays into a lepton pair. Remember: antiparticle arrows point backward in time (downward if time goes up). You must be able to draw Feynman diagrams for beta-minus, beta-plus, electron capture, and electron-proton collisions.

9. The Photoelectric Effect

Photoelectric effect: The emission of electrons from a metal surface when electromagnetic radiation of sufficiently high frequency is incident on it. The emitted electrons are called photoelectrons.

Key Observations

1. No electrons are emitted below a certain **threshold frequency** (f_0), regardless of intensity.

2. Emission is **instantaneous** above the threshold frequency (no time delay).
3. Increasing intensity increases the **number** of photoelectrons emitted per second (the photocurrent), not their maximum kinetic energy.
4. Increasing frequency increases the **maximum kinetic energy** of photoelectrons.
5. Below the threshold frequency, even extremely intense light produces zero photoelectrons.

Why Wave Theory Fails

Classical wave theory predicts that energy is spread continuously across the wavefront. This would mean:

- Any frequency should eventually cause emission if intense enough (wrong).
- There should be a time delay as the electron gradually absorbs energy (wrong).
- Increasing intensity should increase kinetic energy (wrong).

The photoelectric effect provided strong evidence for the particle nature of light.

Photon: A quantum (discrete packet) of electromagnetic energy. Each photon has energy $E = hf = hc/\lambda$, where h is the Planck constant (6.63×10^{-34} J s). Light consists of a stream of photons; intensity is proportional to the number of photons per second.

Einstein's Photoelectric Equation

$$hf = \varphi + E_{k(max)}$$

$$hf = hf_0 + \tfrac{1}{2}mv_{max}^2$$

- **φ** = work function (minimum energy needed to free an electron from the surface), units: J or eV
- **f_0** = threshold frequency: $\varphi = hf_0$
- **$E_{k(max)}$** = maximum kinetic energy of emitted photoelectron
- The equation is one-photon-one-electron: each photon interacts with one electron

Electronvolt (eV): The energy gained by an electron accelerated through a potential difference of 1 V. 1 eV = 1.60×10^{-19} J. This is a convenient unit for atomic-scale energies.

Worked Example: UV light of wavelength 250 nm is incident on a zinc surface with work function 4.3 eV. Calculate whether photoelectrons are emitted and, if so, their maximum kinetic energy.

Photon energy: $E = hc/\lambda = (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / (250 \times 10^{-9})$

$E = 7.96 \times 10^{-19}$ J $= 7.96 \times 10^{-19} / 1.60 \times 10^{-19}$ = **4.97 eV**

Since 4.97 eV > 4.3 eV (work function), photoelectrons ARE emitted.

$E_{k(max)} = hf - \varphi = 4.97 - 4.3$ = **0.67 eV** $= 1.07 \times 10^{-19}$ J

Worked Example: Calculate the threshold frequency for a metal with work function 2.2 eV.

$\varphi = hf_0$

$f_0 = \varphi/h = (2.2 \times 1.60 \times 10^{-19}) / (6.63 \times 10^{-34})$ = **5.31×10^{14} Hz**

This corresponds to green/yellow light ($\lambda = c/f$ = 565 nm).

Exam Tip: A graph of $E_{k(max)}$ vs f is a straight line with gradient = h (Planck constant), x-intercept = f_0 (threshold frequency), and y-intercept = $-\varphi$ (negative work function). Different metals give parallel lines (same gradient h) with different intercepts.

Real-world application: The photoelectric effect is used in solar cells (photovoltaic cells) to generate electricity, in CCD sensors in digital cameras, in photoelectron spectroscopy for surface analysis, and in automatic door sensors.

10. Energy Levels and Photon Emission

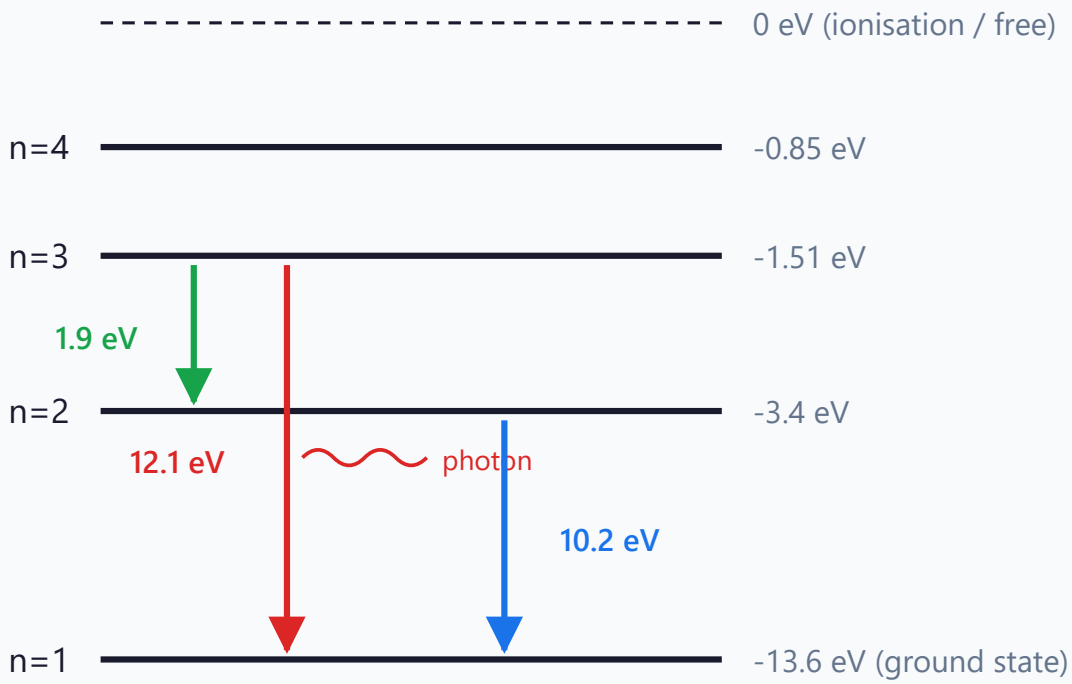
Electrons in atoms occupy discrete energy levels (n = 1, 2, 3...). The ground state (n = 1) has the lowest (most negative) energy. Energy levels are quantised — only certain specific energies are allowed.

Photon Emission

When an electron transitions from a higher energy level (E_2) to a lower one (E_1), a photon is emitted with energy exactly equal to the difference between the two levels:

$$\Delta E = E_2 - E_1 = hf = hc/\lambda$$

Hydrogen Atom Energy Levels



Photon energy = E(higher) - E(lower) = hf = hc/lambda

Worked Example: Calculate the wavelength of a photon emitted when an electron in hydrogen drops from n=3 to n=2.

$\Delta E = E_3 - E_2 = (-1.51) - (-3.4) = 1.89$ eV

Convert to joules: $1.89 \times 1.60 \times 10^{-19} = 3.024 \times 10^{-19}$ J

$\lambda = hc/E = (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / 3.024 \times 10^{-19}$

λ = **6.58×10^{-7} m = 658 nm** (red light — this is the H-alpha line)

Photon Absorption

An atom can absorb a photon **only if** the photon energy exactly matches the difference between two energy levels. The electron is then promoted (excited) to the higher level. If the photon energy does not match any transition, it passes through without being absorbed.

Line Spectra

- **Emission spectrum:** Bright coloured lines on a dark background. Produced when excited atoms de-excite. Each line corresponds to a specific electron transition downward. Different elements produce different patterns of lines.
- **Absorption spectrum:** Dark lines on a continuous spectrum background. Produced when white light passes through a cool gas. Each dark line corresponds to a frequency absorbed by the gas, which matches an emission line.
- **Continuous spectrum:** Produced by hot, dense objects (e.g. a filament lamp, the Sun's core). Contains all wavelengths.

Real-world application: Absorption spectra of starlight reveal the chemical composition of stellar atmospheres. The dark Fraunhofer lines in the solar spectrum show the presence of hydrogen, helium, sodium, calcium, and many other elements. This is how helium was first discovered — in the Sun's spectrum before it was found on Earth.

Ionisation and Excitation

Excitation: When an electron absorbs energy and moves to a higher energy level within the atom. The atom remains neutral. Can be caused by photon absorption or by collisions with other particles (e.g. in a gas discharge tube).

Ionisation: When an electron gains enough energy to escape the atom entirely (reach E = 0 or above). The ionisation energy of hydrogen from the ground state is 13.6 eV.

Exam Tip: In fluorescent tubes, mercury atoms are excited by electron collisions. When they de-excite, they emit UV photons. These UV photons hit the phosphor coating on the inside of the tube, which absorbs the UV and re-emits visible light. This is why the phosphor coating glows.

11. Wave-Particle Duality

Wave-particle duality: Light and matter exhibit both wave-like and particle-like behaviour, depending on the experiment. Light shows wave behaviour in diffraction and interference, and particle behaviour in the photoelectric effect. Electrons show particle behaviour in collisions and wave behaviour in electron diffraction.

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de Broglie Wavelength

Every moving particle has an associated wavelength, proposed by de Broglie in 1924:

$$\lambda = h / p = h / (mv)$$

- λ = de Broglie wavelength (m)
- h = Planck constant (6.63×10^{-34} J s)
- p = momentum (kg m s^{-1})
- m = mass (kg), v = velocity (m s^{-1})

Worked Example: Calculate the de Broglie wavelength of an electron accelerated through a PD of 5.0 kV.

Kinetic energy gained: $E_k = eV = 1.60 \times 10^{-19} \times 5000 = 8.0 \times 10^{-16}$ J

$E_k = ½mv^2$, so $v = \sqrt{(2E_k/m)} = \sqrt{(2 \times 8.0 \times 10^{-16} / 9.109 \times 10^{-31})}$

$v = 4.19 \times 10^7$ m s^{-1}

$\lambda = h/(mv) = 6.63 \times 10^{-34} / (9.109 \times 10^{-31} \times 4.19 \times 10^7)$

λ = **1.74×10^{-11} m = 0.017 nm**

This is smaller than atomic spacing (~0.1 nm), so electron diffraction by crystals is observable.

Evidence for electron waves: In 1927, Davisson and Germer fired electrons at a nickel crystal and observed a diffraction pattern, confirming de Broglie's hypothesis. The spacing of the diffraction maxima matched predictions using $\lambda = h/mv$.

Exam Tip: Larger objects have extremely tiny de Broglie wavelengths. A tennis ball (mass ~0.06 kg, v ~50 m/s) has $\lambda \sim 10^{-34}$ m, far too small to observe any wave effects. Wave-particle duality is only significant at the atomic/subatomic scale where λ is comparable to the size of the objects or gaps involved.

Real-world application: Electron microscopes use the wave nature of electrons. Because high-energy electrons have much shorter wavelengths than visible light, electron microscopes can resolve structures down to ~0.1 nm — about 1000 times better than optical microscopes. This is essential for imaging individual atoms, viruses, and nanostructures.

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